SUPERSTORE

Sales & Profitability Analysis

A SQL Server exploratory data analysis focused on product profitability, regional performance, discounting patterns, customer segments, and order returns.

Business objective

Evaluate Superstore’s sales and profitability to identify underperforming products, regional opportunities, potential discount-related issues, and return patterns. Translate the findings into practical recommendations that can support better business decisions.

Executive summary

The analysis surfaced several areas worth management attention. Tables generated about $208.0K in sales but lost $17.8K, producing a -8.53% profit margin. Machines also showed a thin 1.82% margin despite roughly $189.9K in sales. Higher discount bands were associated with losses in the observed data, while the West region had a notably elevated order return rate of 11.56%, compared with 5.79% overall. These are signals for targeted investigation—not proof of cause and effect.

TABLES PROFIT MARGIN	OVERALL ORDER RETURN RATE	WEST RETURN RATE
-8.53%	**5.79%**	**11.56%**
$208.0K sales; -$17.8K profit	296 of 5,111 distinct orders	Highest among four regions

Scope note: This report summarizes exploratory findings from the Sample Superstore dataset. The data identifies patterns and areas for follow-up; it does not establish the causes of losses or returns.

1. Product profitability: revenue does not guarantee profit

Tables is the clearest warning sign in the subcategory results: it generated $208,020 in sales but recorded a loss of $17,753, resulting in a -8.53% profit margin. Bookcases and Supplies also reported negative margins. Machines generated about $189,925 in sales but only $3,462 in profit, for a 1.82% margin.

Subcategory	Sales	Profit	Margin
Tables	$208,020	-$17,753	-8.53%
Bookcases	$115,361	-$3,632	-3.15%
Supplies	$46,725	-$1,171	-2.51%
Machines	$189,925	$3,462	1.82%
Chairs	$335,768	$27,224	8.11%
Phones	$331,843	$45,051	13.58%
Copiers	$150,745	$56,094	37.21%
Paper	$79,541	$34,512	43.39%

Interpretation: Tables and Bookcases warrant investigation because they combine meaningful sales with losses. Machines deserves attention for a different reason: it is profitable overall, but retains very little profit relative to revenue. In contrast, Copiers and Paper show much stronger margins in this dataset.

Why might Tables be loss-making? The current aggregate results establish the outcome, not its cause. Discounting is one plausible factor, but product mix, underlying costs, and transaction-level patterns may also matter. A useful next step would be to compare discount and profit patterns within Tables and inspect individual products.

2. Customer segments: Consumer contributes the most absolute profit

Segment	Sales	Total profit	Share of total profit
Consumer	$1,170,660	$136,371	46.66%
Corporate	$715,806	$94,250	32.24%
Home Office	$440,068	$61,676	21.10%

Consumer accounts for 46.66% of total profit, with Corporate contributing 32.24%. Together, they account for 78.90% of profit. However, Consumer also has the largest sales volume. Its larger contribution does not necessarily mean it has the highest profit margin; contribution by total profit and profitability per dollar of sales are different measures.

3. Regional performance varies by product

The analysis compared subcategory performance across regions rather than assuming that each product behaves the same everywhere. The working observation from this exploration was that Furnishings performed poorly in the Central region while performing better elsewhere, whereas Tables appeared weak across regions. This suggests that some issues may be region-specific while others may be more widespread.

Evidence limitation: The detailed region-by-subcategory figures for Furnishings and Tables were not included in the summarized results available for this report. Treat this as a lead from the exploration, not a quantified conclusion, until the exact regional sales, profit, and margin figures are captured in the SQL output.

Implication: Compare the same subcategory across regions using total sales, total profit, and profit margin. Prioritize combinations with material sales and negative or unusually low margins for further review.

4. Discounts are strongly associated with lower profitability

Grouping order lines by discount rate revealed a striking pattern in the observed data: every displayed discount level of 30% or higher had negative aggregate profit, while the 10% and 20% groups had positive aggregate profit. The 20% discount group was particularly notable for its substantial total profit.

Discount	Order lines	Total sales	Total profit	Profit margin
80%	301	$16,980	-$30,565	-180.01%
70%	424	$40,805	-$40,301	-98.76%
60%	149	$7,106	-$6,164	-86.75%
50%	66	$58,919	-$20,506	-34.80%
40%	207	$116,498	-$23,087	-19.82%
30%	230	$104,474	-$10,513	-10.06%
20%	3,706	$773,939	$91,080	11.77%
10%	96	$54,952	$9,100	16.56%

The subcategory comparison adds nuance. Tables had a 25.81% average discount and a -8.53% margin; Bookcases had a 21.53% average discount and a -3.15% margin; Machines had a 30.43% average discount and only a 1.82% margin. However, Supplies lost money despite a comparatively low 7.60% average discount, while Binders remained profitable despite a high 36.91% average discount.

Interpretation: Discounting may be contributing to weak profitability, but it does not explain every loss. These are associations, not causal proof. Product mix, cost, and the reasons particular orders received discounts could influence the pattern.

5. Returns: a regional signal stands out

Across 5,111 distinct orders, 296 were marked as returned, giving an order-level return rate of 5.79%. This measures the share of orders returned—not the share of individual units or product lines returned.

Group	Value	Orders	Returned	Return rate
Region	West	1,635	189	11.56%
Region	Central	1,179	39	3.31%
Region	East	1,475	44	2.98%
Region	South	822	24	2.92%
Category	Technology	1,560	123	7.88%
Category	Furniture	1,819	136	7.48%
Category	Office Supplies	3,812	234	6.14%
Segment	Corporate	1,552	93	5.99%
Segment	Consumer	2,628	154	5.86%
Segment	Home Office	931	49	5.26%

The West's return rate of 11.56% is substantially higher than the other regions, which range from 2.92% to 3.31%. This is the clearest return-related signal and deserves follow-up. Technology has the highest category-level rate at 7.88%, while customer-segment rates are relatively close, ranging from 5.26% to 5.99%.

Measurement caveat: The returns file identifies returned order IDs. Category-level rates therefore mean that an order containing that category was associated with a returned order; they do not establish that a specific product in that category was the item returned. Orders can contain multiple categories, so category-level order counts and rates may overlap and should not be summed.

6. Recommendations and next steps

1. Investigate Tables and Bookcases. Review product-level profit, discount rates, and regional patterns to identify whether losses are concentrated in particular products or transactions.

2. Review high-discount transactions. Examine the business rationale and profitability of discounts at 30% and above. Avoid introducing a blanket discount cap until product-level and cost context are understood.

3. Investigate the West's returns. Break returned orders down by product/category, shipping mode, and time period where the data supports it; validate the result and look for operational explanations.

4. Compare absolute profit with profit margin. Consumer contributes the largest share of total profit, but compare segment margins before claiming it is the most efficient segment.

5. Quantify regional product patterns. Capture exact region-by-subcategory sales, profit, and margin results—especially for Furnishings in Central and Tables across regions—before making targeted regional recommendations.

Conclusion

The analysis points to three priorities: investigate loss-making subcategories, review the relationship between discounting and profitability, and understand why the West has a much higher order return rate. These findings provide a practical starting point for deeper analysis; they should guide investigation rather than be treated as causal explanations.

Before publication, confirm that the report's figures match the final SQL outputs and add the exact regional subcategory results if you want to retain the Furnishings/Central observation as a headline finding.